

# A Packing of 9 Four-Spaces in $\mathbb{R}^{13}$

Limin Ge\*

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## Abstract

We give 9 4-dimensional subspaces of  $\mathbb{R}^{13}$  whose smallest squared chordal distance is at least 3.115123. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.076624, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 9 subspaces of dimension 4 in  $\mathbb{R}^{13}$  whose pairwise squared chordal distances  $D(a, b) = 4 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 3.115123\dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 13$ ,  $n = 4$ . The cell  $N = 9$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.076624, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is  $4(13 - 4)/13 \cdot 9/8 = 3.115385$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N9.txt`

in <https://github.com/Ge-limin/math>.

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\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

```
778780843860636010417467504704721659949647416585443618039879322076346633
699319525176517619260312524205532136739710199751488951144800681543554755
000641716943838092054901663130556665972270415743
```

denominator:

```
250000000000102832386555965285462518603089962564920221878102207951602910
207984534938798961712228874688155049948561458793626648743698039839577914
821410553725881181303235910507243970737500004688
```

## B Coordinates

9 blocks of 13 rows with 4 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m13n4N9.txt`.

```
-303339634515 369712744453 -27894221971 -232814276194 5948426210 -181527539654 331369306597 -597303343374 6611054722 -128622453197 -17801944110 -247484844988
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-104736914134 -260687874951 16388376765 603114055540 6537624215 81256235702 -136619534499 568652415432 48477456326 121967595151 372275268733 -253841520123
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```